

AB RENEWABLE ENERGY ENGINEERING

Engr. Alexis T. Belonio, MS, ABE, ASEAN Engineer
*Freelance Consultant and Technology Developer; Former
Senior Research Fellow (PhilRice) and
Associate Professor (Central Philippine University)
Member: PSABE, ASABE, ASME*

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BIOETHANOL FERMENTATION AND DISTILLATION

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1. Fuel derived from biomass primarily used for motive, thermal, and power generation.

- a. Bio-Oil
- b. Biogas
- c. Biofuel
- d. All of the above

2. Biofuel, with chemical formula C_2H_5OH , produced from biomass feedstock.

- a. Bioethanol
- b. Biodiesel
- c. Bio-oil
- d. All of the above

3. Process of producing bioethanol from sugar-rich plant materials.

- a. Anaerobic digestion
- b. Liquefaction
- c. Fermentation and distillation
- d. None of the above

4. Organism responsible for the fermentation of sugar into bioethanol.

- a. Bacteria
- b. Yeast
- c. Fungi
- d. None of the above

5. A 10,000liter-per-day-capacity bioethanol plant is using fermented sap from sweet sorghum with 15% alcohol. If the plant has 90% distillation efficiency to produce anhydrous bioethanol, what is the volume of ethanol that can be derived from the plant in one day?

- a. 967 liters per day
- b. 1,250 liters per day
- c. 1,350 liters per day
- d. None of the above

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6. How much stillage can be derived from the bioethanol plant in Item 5?

- a. 8,650 liters per day
- b. 9,247 liters per day
- c. 9,950 liters per day
- d. None of the above

7. Product resulting from subjecting sugar-rich biomass to fermentation and distillation.

- a. Producer gas
- b. Biodiesel
- c. Bioethanol
- d. Biogas
- e. None of the above

8. Bioethanol is produced by subjecting ____ to fermentation and distillation.

- a. coconut oil
- b. coconut sap
- c. coconut water
- d. None of the above

9. Average heating value of bioethanol is ____.

- a. 3,000 kcal/liter
- b. 5,130 kcal/liter
- c. 8,250 kcal/liter
- d. None of the above

10. Substance used as catalyst in producing alcohol from fermented sap of sugar-rich plant material.

- a. Yeast
- b. Bacteria
- c. Mold
- d. All of the above

11. Microorganism responsible for producing bioethanol from feedstock.

- a. Bacteria
- b. Mold
- c. Yeast
- d. All of the above

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12. Product of distillation which is composed of water and alcohol.

- a. Stillage
- ☒ b. Distillate
- c. Residue
- d. None of the above

13. Microorganism responsible in the decomposition of biomass which does not produce alcohol or acid.

- ☒ a. Bacteria
- b. Mold
- c. Yeast
- d. None of the above

14. Microorganism responsible for producing acid from feedstock instead of alcohol.

- ☒ a. Bacteria
- b. Mold
- c. Yeast
- d. None of the above

15. Ratio of the volume of alcohol obtained from the distillate to the volume of alcohol processed from the fermented feedstock multiplied by 100.

- a. Alcohol recovery
- b. Alcohol content
- ☒ c. Distillation efficiency
- d. None of the above

16. A 1000ml sample of feedstock contains 28.1% alcohol. After distillation, the volume of distillate recovered was 313ml with 66.9% alcohol content. What is the distillation efficiency?

- ☒ a. 45.3%
- ☒ b. 74.5%
- c. 86.9%
- d. None of the above

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17. Yeast requirement for fermenting nipa sap for the production of bioethanol.

- a. 0.1 g of yeast per liter of sap
- b. 100 g of yeast per m³ of sap
- ☒ c. ¼ teaspoon per liter of sap
- d. All of the above
- e. None of the above

18. Liquid that remains from the "still" or from a boiler of the bioethanol plant distillation unit.

- ☒ a. Stillage
- b. Distillate
- c. Residue
- d. None of the above

19. Percentage ratio of the volume of alcohol produced from the feedstock after subjecting it in a "still" to the alcohol present in the input material.

- ☒ a. Distillation efficiency
- b. Conversion efficiency
- c. Distillation ratio
- d. All of the above

20. A 200liter-capacity bioethanol distiller contains a feedstock with 10% alcohol content. After 2 hours of distillation process, it produces 30 liters of crude bioethanol with 35% alcohol content. What is the alcohol recovery of the distiller?

- ☒ a. 30.5%
- ☒ b. 52.5 %
- c. 60.2 %
- d. None of the above

21. Referring to Item 20, how much stillage remained at the distiller?

- ☒ a. 140 liters
- ☒ b. 170 liters
- c. 180 liters
- d. None of the above

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GOD BLESS !!!

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